

Suchetan Karloopia

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Professional Summary

Final Year **B.Tech Chemical Engineering** student (NIT Srinagar) with hands-on exposure to **ammonia/urea production, process simulation (COMSOL, Aspen Plus, MATLAB), and P&ID interpretation**. Skilled in heat and mass transfer, reaction engineering, and data-driven process optimization. Passionate about **energy efficiency, sustainability, and large-scale industrial operations**.

Education

National Institute Of Technology, Srinagar, B.Tech. in Chemical Engineering July 2022 – Aug 2026

- CGPA: 6.73/10.0 up to 7th Semester (View Transcript)
- **Coursework:** Chemical Process Principles, Thermodynamics, Fluid Mechanics, Heat and Mass Transfer, Process Control, Process Equipment Design, Chemical Reaction Engineering, Energy Technology, Process Safety, Biochemical Engineering, Process Economics and Plant Design.
- **Practical Work In:** Mechanical Operations, Fluid Mechanics (Venturi/orifice meter), Heat Transfer (heat exchangers, convection), Computer Simulation (COMSOL, Excel), Thermodynamics & Reaction Engineering (PFR, CSTR, Batch), Mass Transfer (packed bed, cooling tower), Process Dynamics & Control (level, flow, temp., pressure trainers).

Kendriya Vidyalayan Sunjuwan, Jammu, Higher Senior Secondary in Non-Medical Apr 2021 – March 2022

- Percentage: 73.8 **Coursework:** Physics, Chemistry, Mathematics, English and Computer Science.

Experience

Vocational Trainee, National Fertilizers Limited (NFL) - Nangal Unit, Punjab, India Jan 2025 - Feb 2025

(View Certificate)

- Worked in the **Production Department** - Ammonia and Urea Plant.
- Observed large-scale production processes for ammonia and urea, including steam reforming, CO₂ removal, and synthesis loop operations.
- Gained knowledge of PFD and P & ID interpretation, safety protocols, and unit operation control in a real plant environment.
- Developed understanding of **Process efficiency** and **energy optimization** in fertilizer manufacturing.

Intern, IICChE Training Institute (IICChE-TI) – Online Internship Programme Feb 2026 – Apr 2026

(View Certificate)

- Completed a **6-week programme** on **MATLAB and AI/ML for Chemical Process Industries**, covering process simulation, data analysis, and predictive modeling.
- Applied machine learning techniques to chemical engineering problems including process optimization and intelligent control systems.

Key Projects and Practical Exposure

- **Final Year Project (B.Tech Thesis):** *Development and Characterization of Bio-Composites from Waste Aquatic Biomass for Sustainable Insulation* - Supervised by Dr. Fasil Qayoom Mir, HOD and Associate Prof., Department of Chemical Engg., NIT Srinagar. Fabricated four composite panels (S1-S4) from water lily (Dal Lake) with varying particle sizes and corn starch binder ratios; completed ASTM characterization on moisture content, bulk density, water absorption, mechanical, SEM and thermal conductivity testing. Achieving Insulation Grade Results.
- **COMSOL Multiphysics Software:** Modeled turbulent water flow in serpentine geometry (Re = 5000-10000) and also made for two immiscible liquids and studied different flows in different geometries. (View Project models)
- **Process Equipment Handling:** Operated lab-scale heat exchangers, reactors (PFR, CSTR, packed bed etc.), mass transfer equipment and mechanical operating units.

Skills

Technical & Professional	Computer Skills	Languages
• Chemical Process Design • Heat and Mass Transfer • Chemical Reaction Engineering • Process Safety & Environment • Team Collaboration & Leadership • Project Management • Bioresource Technology	• Aspen Plus Software • MATLAB Software • Microsoft Office (Excel, Word, PowerPoint) • COMSOL Multiphysics • Python and MySQL	• English: Fluent • Hindi: Native • German: Beginner (A1)

Certifications And Achievements

National Level Speed Skating Championship In Bhopal, India: secured Bronze in 100m time trial for my school Kendriya Vidyalayan Sunjuwan, Jammu.

Active Member: AIChE - USA and IICChE - India

Short term Course On: Recent trends and application of Nanotechnology in Engineering. (View Certificate)